Canonical Quantization of Scalar Fields

Before we go back to the hamiltonian from section 1, let us set

$$c = \hbar = 1$$

for our convenience. Those factors could be reattained using dimensional analysis if needed. Now let us go back to previously constructed hamiltonian. For the case of free particles (no external potential working on the particle),

$$H = \int \mathrm{d}^3x a^\dagger(\boldsymbol{x})\left(-\frac{1}{2m}\nabla^2\right)a(\boldsymbol{x}) \ .$$

Using inverse Fourier transformation on the integral, we can set the integral on momentum space,

$$a(\boldsymbol{x}) = \int \frac{\mathrm{d}^3p'}{(2\pi)^{\frac{3}{2}}}\exp(-i\boldsymbol{p}'\cdot\boldsymbol{x})\tilde{a}(\boldsymbol{p}') \ .$$

So, substituting back to the previous hamiltonian,

$$\begin{aligned}
H &= \int \mathrm{d}^3x\,\mathrm{d}^3p''\,\mathrm{d}^3p'\frac{1}{(2\pi)^3}\exp(i\boldsymbol{p}''\cdot\boldsymbol{x})\tilde{a}^\dagger(\boldsymbol{p}'')\left(-\frac{1}{2m}\nabla^2\right)\exp(-i\boldsymbol{p}'\cdot\boldsymbol{x})\tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3x\,\mathrm{d}^3p''\,\mathrm{d}^3p'\frac{1}{(2\pi)^3}\exp(i\boldsymbol{p}''\cdot\boldsymbol{x})\tilde{a}^\dagger(\boldsymbol{p}'')\left(-\frac{1}{2m}(-i\boldsymbol{p}')^2\right)\exp(-i\boldsymbol{p}'\cdot\boldsymbol{x})\tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3p''\,\mathrm{d}^3p'\,\mathrm{d}^3x\frac{1}{(2\pi)^3}\exp(i(\boldsymbol{p}''-\boldsymbol{p}')\cdot\boldsymbol{x})\frac{1}{2m}(\boldsymbol{p}')^2\tilde{a}^\dagger(\boldsymbol{p}'')\exp(-i\boldsymbol{p}'\cdot\boldsymbol{x})\tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3p''\,\mathrm{d}^3p'\,\delta(\boldsymbol{p}''-\boldsymbol{p}')\frac{1}{2m}(\boldsymbol{p}')^2\tilde{a}^\dagger(\boldsymbol{p}'')\tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3p\frac{1}{2m}\boldsymbol{p}^2\tilde{a}^\dagger(\boldsymbol{p})\tilde{a}(\boldsymbol{p}) \ .
\end{aligned}$$

The commutation (anticommutation) relations for the operators are

$$\begin{aligned}
\left[\tilde{a}(\boldsymbol{p}),\tilde{a}(\boldsymbol{p}')\right]_\mp &= 0 \\
\left[\tilde{a}^\dagger(\boldsymbol{p}),\tilde{a}^\dagger(\boldsymbol{p}')\right]_\mp &= 0 \\
\left[\tilde{a}(\boldsymbol{p}),\tilde{a}^\dagger(\boldsymbol{p}')\right]_\mp &= \delta(\boldsymbol{p}-\boldsymbol{p}')
\end{aligned}$$

Like the annhilation operator,

$$\tilde{a}(\boldsymbol{p})|0\rangle = 0 \ .$$

We could see from the hamiltonian that energy is defined as $E(\boldsymbol{p}) = \frac{\boldsymbol{p}^2}{2m}$. If we want to generalize it to the relativistic limits, we could set the energy using the energy-momentum relation, that is $E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2c^2 + m^2c^4}$. Substituting the energy back into the hamiltonian,

$$H = \int \mathrm{d}^3p\sqrt{\boldsymbol{p}^2c^2 + m^2c^4}\tilde{a}^\dagger(\boldsymbol{p})\tilde{a}(\boldsymbol{p}) \ .$$

This is the relativistic hamiltonian for free spin-0 particles (both bosons and fermions). To see if the theory is Lorentz invariant, let us go back to the Klein-Gordon equation, that is the Lorentz invariant equation of motion for classical scalar field.

$$(-\partial^2 + m^2)\varphi(x) = 0 \ .$$

Because φ is only classical field, the m is not yet to be mass as its dimension is $[L]^{-1}$. This equation of motion can be derived from the action principle $S = \int \mathrm{d}^4x\mathcal{L}$, where $L = \int \mathrm{d}^3x\mathcal{L}$ is the integral of lagrangian density $\mathcal{L}$. Action on another frame would be $\bar{S} = \int \mathrm{d}^4x\bar{\mathcal{L}}$. Let us take a look at Lorentz transformation.

$$\begin{aligned}
\bar{x}^\mu &= \Lambda^\mu{}_\nu x^\nu \\
\partial_\alpha \bar{x}^\mu &= \Lambda^\mu{}_\nu \partial_\alpha x^\nu \\
\partial_\alpha \bar{x}^\mu &= \Lambda^\mu{}_\nu \delta_\alpha{}^\nu \\
\Lambda^\mu{}_\nu &= \partial_\nu \bar{x}^\mu \\
&= \frac{\partial \bar{x}^\mu}{\partial x^\nu} \ .
\end{aligned}$$

We can see that the Lorentz matrix also acts as the Jacobian between two inertial frame. So, the infinitesimal volume of the two frames are related with

$$\begin{aligned}
\mathrm{d}^4\bar{x} &= \det(J)\,\mathrm{d}^4x \\
&= \det(\Lambda)\,\mathrm{d}^4x \\
&= \mathrm{d}^4x \ .
\end{aligned}$$

Since we are taking consideration only of proper orthochronous transformation, infinitesimal volumes are the same between two frames. Now, if we want to make sure that action S is invariant, the only left is the lagrangian density. We then set that the lagrangian density must be the same in two frames

$$\bar{\mathcal{L}} = \mathcal{L} \ ,$$

so now,

$$\bar{S} = \int \mathrm{d}^4\bar{x}\bar{\mathcal{L}} = \int \mathrm{d}^4x\mathcal{L} = S \ .$$

Let us take the lagrangian density as

$$\mathcal{L} = -\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2 \ .$$

If we take the variation of action based on that lagrangian and apply the least action principle,

$$\begin{aligned}
\delta S &= 0 \\
\int \mathrm{d}^4x\,\delta\mathcal{L} &= 0 \\
\int \mathrm{d}^4x\left[-\frac{1}{2}\partial^\mu\,\delta\varphi\partial_\mu\varphi - \frac{1}{2}\partial^\mu\varphi\partial_\mu\,\delta\varphi - \frac{1}{2}m^2(2\varphi\,\delta\varphi)\right] &= 0 \\
\int \mathrm{d}^4x\left[-\frac{1}{2}\partial^\mu\,\delta\varphi\partial_\mu\varphi - \frac{1}{2}\partial^\mu\varphi\partial_\mu\,\delta\varphi - m^2\varphi\,\delta\varphi\right] &= 0 \\
-\frac{1}{2}\left[\big(\partial_\mu\varphi + \partial^\mu\varphi\big)(\delta\varphi)|_{-\infty}^{\infty} - 2\int \mathrm{d}^4x\,\delta\varphi\partial^\mu\partial_\mu\varphi\right] - \int \mathrm{d}^4x m^2\varphi\,\delta\varphi &= 0 \\
\int \mathrm{d}^4x\big(\partial^\mu\partial_\mu - m^2\big)\varphi\,\delta\varphi &= 0 \ ,
\end{aligned}$$

so we obtained Klein-Gordon equation from the invariant lagrangian density. The general solution for Klein-Gordon is

$$\varphi(\boldsymbol{x},t)=\int\frac{\mathrm{d}^3k}{f(k)}\left[a(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}-i\omega t}+b(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}+i\omega t}\right]\ .$$

We must impose that $\varphi(x)=\varphi^*(x)$ to ensure that the field equation is real.

$$\begin{aligned}\varphi^*(\boldsymbol{x},t)&=\int\frac{\mathrm{d}^3k}{f(k)}\left[a^*(\boldsymbol{k})e^{-i\boldsymbol{k}\cdot\boldsymbol{x}+i\omega t}+b^*(\boldsymbol{k})e^{-i\boldsymbol{k}\cdot\boldsymbol{x}-i\omega t}\right]\\&=\int\frac{\mathrm{d}^3k}{f(k)}\left[b^*(-\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}-i\omega t}+a^*(-\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}+i\omega t}\right]\ .\end{aligned}$$

Equating both fields, we get that $a(\boldsymbol{k})=b^*(-\boldsymbol{k})$ and $b(\boldsymbol{k})=a^*(-\boldsymbol{k})$. So, changing all coefficient to a, the field equation becomes

$$\begin{aligned}\varphi(\boldsymbol{x},t)&=\int\frac{\mathrm{d}^3k}{f(k)}\left[a(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}-i\omega t}+a^*(-\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}+i\omega t}\right]\\&=\int\frac{\mathrm{d}^3k}{f(k)}\left[a(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{x}-i\omega t}+a^*(\boldsymbol{k})e^{-i\boldsymbol{k}\cdot\boldsymbol{x}+i\omega t}\right]\ .\end{aligned}$$

By defining 4-vector $k^\mu=(\omega,\boldsymbol{k})$,

$$\varphi(x)=\int\frac{\mathrm{d}^3k}{f(k)}\left[a(\boldsymbol{k})e^{ik^\mu x_\mu}+a^*(\boldsymbol{k})e^{-ik^\mu x_\mu}\right]\ .$$

We define **on-shell** momentum as $k^2=-m^2$. So, in this case,

$$k^2=-\omega^2+\boldsymbol{k}^2=-m^2\ ,$$

if we take $\exp i(\boldsymbol{k}\cdot\boldsymbol{x}\pm\omega t)$ as one solution of the Klein-Gordon equation.

Now we move on to choose the factor of $\frac{1}{f(k)}$ inside the integrand. An integration measure that is Lorentz (proper-orthochronous) invariant is $\int\mathrm{d}k^0\,\delta(k^2+m^2)\theta(k^0)$ where δ and θ are the delta Dirac and step function.

$$\begin{aligned}\int_{-\infty}^{\infty}\mathrm{d}k^0\,\delta(k^2+m^2)\theta(k^0)&=\int_{-\infty}^{\infty}\mathrm{d}k^0\,\delta\Big(\boldsymbol{k}^2+m^2-\left(k^0\right)^2\Big)\theta(k^0)\\&=\int_{-\infty}^{\infty}\mathrm{d}k^0\,\delta\Big(\omega^2-\left(k^0\right)^2\Big)\theta(k^0)\\&=\int_{-\infty}^{\infty}\mathrm{d}k^0\,\delta(g(k^0))\theta(k^0)\\&=\sum_i\frac{\theta({k^0}_i)}{|g'({k^0}_i)|}\\&=\frac{1}{2\omega}\ .\end{aligned}$$

Then we can take

$$\frac{1}{f(k)}\equiv\frac{1}{(2\pi)^3 2\omega}$$

and also introduce a new momentum differential

$$\widetilde{dk} \equiv \frac{\mathrm{d}^3 k}{(2\pi)^3 2\omega} \ .$$

Substituting both new quantities into the field equation, we would get the form

$$\begin{aligned}
\varphi(x) &= \int \widetilde{\mathrm{d}k}\left[a(\boldsymbol{k})e^{ik^\mu x_\mu} + a^*(\boldsymbol{k})e^{-ik^\mu x_\mu}\right] \\
\int \mathrm{d}^3x e^{-ik'x}\varphi(x) &= \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x a(\boldsymbol{k})e^{i(k-k')x} + \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x a^*(\boldsymbol{k})e^{-i(k+k')x} \\
&= \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x e^{i(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{x}} e^{i(\omega'-\omega)t}a(\boldsymbol{k}) + \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x e^{-i(\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{x}} e^{i(\omega'+\omega)t}a^*(\boldsymbol{k}) \\
&= \int \frac{\mathrm{d}^3k}{2\omega}\,\delta(\boldsymbol{k}-\boldsymbol{k}')e^{i(\omega'-\omega)t}a(\boldsymbol{k}) + \int \frac{\mathrm{d}^3k}{2\omega}\,\delta(\boldsymbol{k}'+\boldsymbol{k})e^{i(\omega'+\omega)t}a^*(\boldsymbol{k}) \\
&= \frac{1}{2\omega'}a(\boldsymbol{k}') + \frac{\exp(2i\omega' t)}{2\omega'}a^*(-\boldsymbol{k}') \ .
\end{aligned}$$

Because we have the on-shell condition, that is $\omega = +\sqrt{\boldsymbol{k}^2 + m^2}$, the integration over 3D delta Dirac also forces $\omega' = \omega$. Changing the parameter again,

$$\int \mathrm{d}^3x e^{-ikx}\varphi(x) = \frac{1}{2\omega}a(\boldsymbol{k}) + \frac{\exp(2i\omega t)}{2\omega}a^*(-\boldsymbol{k}) \ .$$

Let us take the 0-th derivative of field equation.

$$\begin{aligned}
\partial_0\varphi(x) &= i\omega\int \widetilde{\mathrm{d}k}\left[a(\boldsymbol{k})e^{ik^\mu x_\mu} - a^*(\boldsymbol{k})e^{-ik^\mu x_\mu}\right] \\
\int \mathrm{d}^3x e^{-ikx}\partial_0\varphi(x) &= \frac{i}{2}a(\boldsymbol{k}) + \frac{i\exp(2i\omega t)}{2}a(-\boldsymbol{k}) \ .
\end{aligned}$$

Now, we can combine both equation in order to express the coefficient a.

$$\begin{aligned}
a(\boldsymbol{k}) &= \int \mathrm{d}^3x \exp(-ikx)[i\partial_0\varphi(x) + \omega\varphi(x)], \\
a^\dagger(\boldsymbol{k}) &= \int \mathrm{d}^3x \exp(ikx)[-i\partial_0\varphi(x) + \omega\varphi(x)].
\end{aligned}$$

We now can start generalizing the hamiltonian. Remember that the classical hamiltonian is

$$\begin{aligned}
H &= \sum_i p_i\dot{q}_i - L, \\
p_i &= \frac{\partial L}{\partial \dot{q}},
\end{aligned}$$

where p is the conjugate momentum and q is the generalized coordinates. The proper generalization to field theory would be that the continuous q is replaced by $\varphi(x)$ since our now lagrangian density is explicitly dependent on φ and its derivative. Then we have hamiltonian density expressed as

$$\begin{aligned}
\mathcal{H} &= \Pi\dot{\varphi} - \mathcal{L}, \\
\Pi &= \frac{\partial \mathcal{L}}{\partial \dot{\varphi}}.
\end{aligned}$$

Let us find the hamiltonian from the previous lagrangian.

$$
\begin{aligned}
\Pi &= \frac{\partial}{\partial \dot{\varphi}}\left(-\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2\right) \\
&= \frac{\partial}{\partial \dot{\varphi}}\left(\frac{1}{2}\dot{\varphi}^2\right) \\
&= \dot{\varphi}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{H} &= \dot{\varphi}^2 + \frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi + \frac{1}{2}m^2\varphi^2 - \Omega_0 \\
&= \dot{\varphi}^2 - \frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}(\boldsymbol{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2 - \Omega_0 \\
&= \frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}(\boldsymbol{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2 - \Omega_0.
\end{aligned}
$$

Integrating over volume, we could get the hamiltonian in terms of $a(\boldsymbol{k})$.

$$
\begin{aligned}
H &= \int \mathrm{d}^3x\mathcal{H} \\
&= -\Omega_0 + \frac{1}{2}\int \widetilde{\mathrm{d}k}\omega[a^*(\boldsymbol{k})a(\boldsymbol{k}) + a(\boldsymbol{k})]
\end{aligned}
$$